Fast Gradient Boosting without Trees for Large-Scale Time Series Classification

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Abstract

On large time series datasets, training the classifier becomes increasingly expensive. A boosted tree ensemble such as XGBoost searches every feature for splits in every round, which means its cost grows with the size of the data and the whole training set has to sit in memory. We replace the tree with a *global partition*, a handful of threshold predicates that index a table of leaf values, with each threshold placed between a pair of hard, differently labelled examples. The resulting booster, HashBoost, trains from mini-batches on the GPU. Its cost depends on how many rounds we give it rather than on how much data there is, and its memory use does not grow with the training set. We evaluate it on MONSTER, a benchmark of 26 large time series datasets, using QUANT features. With both models tuned by the same procedure and scored on the official test folds, HashBoost matches tuned XGBoost's accuracy and trains 8.2× faster in total on the same GPU. On the larger datasets, where XGBoost is most expensive, it is up to 31× faster, although it is slower on the cheapest ones. The advantage disappears on the small datasets of the UCR archive, where QUANT's own ExtraTrees classifier is both faster and more accurate.

1 Introduction

Gradient boosted decision trees [1], in implementations such as XGBoost [2], LightGBM [3] and CatBoost [4], are the default choice for tabular prediction, where they remain competitive with, and frequently superior to, deep learning [5, 6]. Tree ensembles are also the usual classifier on top of interval-based time series features: QUANT [7] passes its quantile features to extremely randomised trees, and PULSAR [8] is used the same way.

At the scale of the MONSTER benchmark [9], with up to 5.9 million series per dataset and tens of thousands of features after QUANT, the classifier becomes the expensive part. To grow a boosted tree, the learner searches for split points on every feature, and a multiclass booster grows one tree per class in every round. The cost of a round is therefore proportional to the number of rows, features and classes multiplied together. Histogram binning, approximate split finding, gradient-based sampling and feature bundling [2, 3, 10] all reduce the constant factor, but every round still has to revisit the whole training set, and the training set has to be held in memory.

In this paper we replace each tree with a *global partition* of the feature space. A partition is b axis-aligned threshold predicates, applied to every example at once, and their b bits index a table of 2^b leaf values. Each predicate takes a random feature and puts its threshold halfway between two hard examples with different labels, so no search over the training set is needed. The partition also does not depend on which examples happen to be in the current batch. Its leaf values can be re-estimated from any later mini-batch, and the model never needs the whole training set in memory at once.

We claim that this booster matches tuned XGBoost's accuracy on large-scale time series classification. Because its cost does not grow with the data, it also trains far faster, and the gap is largest on exactly the datasets where XGBoost is most expensive. We ask:

RQ1 Given the same tuning effort, does HashBoost match a tuned boosted tree ensemble's accuracy on MONSTER's test folds?

RQ2 How much faster is HashBoost, on which datasets, and why? What does each model reach in the same training time, and what memory does each need?

RQ3 Where does the advantage stop? Does it hold on small training sets?

Contributions.

1. **HashBoost**, a gradient booster whose weak learner is a global partition, with a GPU implementation that trains from mini-batches (Section 3).
2. **A test-fold comparison on all 26 MONSTER datasets** against XGBoost tuned by the same procedure and against QUANT's ExtraTrees, trained on whole pools of up to 4.65 million rows (Section 5.1).
3. **An account of where the time goes**: per-dataset speedups explained by XGBoost's cost per round, a time-matched comparison, and memory against training-set size (Section 5.2).
4. **The small-data boundary** on the 112 UCR datasets (Section 5.3).

Our code is available at `https://github.com/minip8/FIT2082`.

2 Background and Related Work

Boosting. Boosting combines weak learners into a strong one. AdaBoost [11] formalised the idea, and gradient boosting [1] generalised it to any differentiable loss. Trees are the usual weak learner, although nothing in the framework requires them. Modern implementations fit each tree to a second-order expansion of the loss, which makes a leaf's value a Newton step $-\sum g / \sum h$ over its examples [2]. HashBoost keeps this leaf rule and changes only how examples are assigned to leaves.

Oblivious trees and decision tables. CatBoost [4] grows *oblivious* trees, in which every node at a given depth uses the same split. A depth-d tree is then d predicates indexing 2^d leaves, which is exactly the structure of one HashBoost round. The closest relative is boosted decision tables [12]. Like CatBoost, they choose each test greedily by gain over the whole training set (and then *backfit* them, re-choosing each test given the others), and they set their leaves once, without revisiting earlier tables. HashBoost differs on both counts. Its tests are never searched for over the training set, and each new batch re-estimates the leaves of every earlier round. In that respect it is closer to backfitting in additive models [13], carried out online.

Randomised partitions. Random forests [14] and extremely randomised trees [15] showed that much of a tree's split search can be replaced by randomness, as long as the ensemble is large. HashBoost's thresholds are also random, but they are informed by the current model. Each one sits between two hard examples of different classes, so each bit tends to separate examples that the model is currently confusing.

Scaling and streaming. The usual way to make tree boosting fast at scale is to keep the search and speed it up. GPU histogram construction [16] builds each tree's histograms in parallel, and out-of-core training [2] pages the binned matrix in from disk. Both still pass over the whole training set in every round. The other route is to learn from a stream. Hoeffding trees [17] grow a single tree from a stream, splitting a leaf once enough examples have arrived, and online boosting [18, 19] updates every weak learner as each example arrives. HashBoost also updates all of its existing learners from each batch, as online boosting does. Their structure, though, is fixed when they are created, so an update is a scatter-add rather than a refit.

Time series classification. Random-feature methods have shown that data-dependent search can often be replaced by cheap random transformations. ROCKET [20], MiniRocket [21] and Hydra [22] pair random convolutional kernels with a linear classifier, at a fraction of the cost of ensembles such as HIVE-COTE [23]. Interval methods (Time Series Forest [24], CIF [25], DrCIF [23], rSTSF [26], QUANT [7], PULSAR [8]) all finish with some kind of tree. We keep QUANT as a fixed transform and replace only the classifier.

3 Method

3.1 A round is a global partition

Let $x \in \mathbb{R}^p$ be a feature vector and k the number of classes. A round r is defined by b pairs $(j_{r,i}, t_{r,i})$ of a feature index and a threshold, for $i = 0, \dots, b-1$. It maps x to a b-bit code

$$c_r(x) = \sum_{i=0}^{b-1} 2^i \mathbb{1}\big[x_{j_{r,i}} \le t_{r,i}\big] \in \{0, \dots, 2^b - 1\}, \tag{1}$$

and has a table $L_r \in \mathbb{R}^{2^b \times k}$ of leaf values. The model after R rounds is

$$F(x) = \sum_{r=1}^{R} L_r[c_r(x)] \in \mathbb{R}^k, \tag{2}$$

and predicts $\mathrm{softmax}(F(x))$. In other words, a round is an oblivious tree of depth b whose splits are fixed when it is created. By default $b = 8$, which gives each round 256 buckets.

3.2 Choosing the predicates

Given the current batch, its labels and the model's predicted probabilities, a new round's predicates are chosen as follows:

1. Order the batch by cross entropy, hardest first.
2. Walk that order and greedily form b pairs of examples with different labels.
3. For each pair (u, v), draw a feature j uniformly at random and set $t = (u_j + v_j)/2$.

Each bit separates two examples that the model currently gets wrong, and the only search involved is the sort in the first step. We also use a *gain-selected* variant. It draws a pool of bK hard-pair candidates and fills the bits one at a time, much as CatBoost grows an oblivious tree level by level (Section 2). Each bit takes the unused candidate with the highest second-order gain, $\sum G^2/(H + \lambda)$ with $\lambda = 1$, summed over classes and over the cells made by the earlier bits, where G and H are the batch's gradient and hessian summed within a cell. The difference from CatBoost is that HashBoost scores only these bK random candidates, on a single batch, instead of every feature over the whole training set.

Algorithm 1 One HashBoost step on a mini-batch (X, y) of n examples, adding M rounds.

for $m = 1, \dots, M$ **do**
  $F \leftarrow \sum_{r=1}^{R} L_r[c_r(X)]$ ▷ (n, k) logits
  $p \leftarrow \mathrm{softmax}(F)$; $g \leftarrow p - \mathrm{onehot}(y)$; $h \leftarrow p(1-p) + 10^{-3}$
  choose predicates of round $R+1$ from hard pairs in (X, y, p)
  for $r = 1, \dots, R+1$ **do** ▷ all rounds, including the new one
    $G_r[c_r(X)] \mathrel{+}= -g$; $H_r[c_r(X)] \mathrel{+}= h$ ▷ scatter-add
    $L_r \leftarrow \eta\, G_r/(H_r + \epsilon)$
  end for
  $R \leftarrow R + 1$
end for

3.3 Fitting the leaves

We use the softmax cross-entropy gradient $g = p - \mathrm{onehot}(y)$ and a floored diagonal hessian $h = p(1-p) + 10^{-3}$. Every round keeps running sums $G_r, H_r \in \mathbb{R}^{2^b \times k}$ of $-g$ and h over the examples that landed in each bucket, and its leaf values are

$$L_r[s] = \eta \, \frac{G_r[s]}{H_r[s] + \epsilon}, \tag{3}$$

with learning rate $\eta = 0.1$. By default $\epsilon = 10^{-6}$; a larger ϵ is an L2 penalty that shrinks thinly populated leaves.

The main departure from tree boosting is *when* these sums are accumulated. In traditional boosting, a tree is frozen once it is fitted. In HashBoost, every mini-batch is accumulated into every existing round, with gradients evaluated at the current model (Algorithm 1). A round created early keeps absorbing evidence from all the batches that follow, so it is not tied to the batch that created it, and the model can learn from a stream. Adding $M > 1$ rounds per batch raises capacity without needing more passes over the data.

3.4 What training costs

A batch of n rows costs one encoding and one scatter-add for each round it touches, $O(nRb + nRk)$, plus the cost of choosing its new predicates from that batch alone. None of this depends on the size N of the training pool. The total cost is the number of batches times the cost per batch, and memory only has to hold one batch and the tables. A boosted tree's round, by contrast, builds feature histograms over all N rows for each of the k class trees. Its cost per round grows with $N \cdot p \cdot k$, and it needs the binned matrix of all $N \times p$ values.

HashBoost does have a scaling problem of its own. Since every batch updates every round, a batch costs $O(R)$ once there are R rounds, and T batches cost $O(T^2)$. *Round freezing* removes this growth. With a window of W rounds, only the newest W keep accumulating, and once a round is frozen its contribution to an example never changes. Given stable row ids, we can sum each row's frozen contribution once and cache it in an (N, k) table. A batch then costs $O(W + f)$, where f is the number of rounds frozen since its rows were last seen, and the total cost becomes linear in the number of rounds.

3.5 Implementation

HashBoost runs entirely on the GPU in PyTorch. We first wrote a CPU prototype in numba and kept it as a test oracle while porting to the GPU. The GPU version is about 13× faster.

The code is organised so that an experiment can change one component at a time. The predicate family, the split selector and the loss are separate interfaces (Python `Protocol`s). Every experimental change is an opt-in option, which keeps the default model identical to the

prototype's algorithm, and every run writes its parameters and results to disk. We also ported the QUANT transform to PyTorch, so the features are computed on the GPU, right next to the model that trains on them.

4 Experimental Setup

MONSTER. We use all 26 classification datasets in MONSTER [9] (Table 1), with QUANT [7] features (depth 6, div = 4, our PyTorch port). MONSTER supplies five fixed test folds, and for each fold we call every series outside it the *pool*. A fixed shuffle of the pool (seed 42) gives a validation slice and a tune slice of 4,096 rows each, or a fifth of the pool each when the pool has fewer than 73,728 rows. Every model trains on the rest. The tune slice is used for every choice made after training, such as early stopping, and validation error is used only to choose between configurations.

Configurations. HashBoost and XGBoost were tuned in the same way, using only fold 0's validation error. We screened one change at a time on 10 of the 26 datasets (CrowdSourced, DREAMERA, FordChallenge, InsectSound, LenDB, PAMAP2, STEW, Traffic, UCIActivity and WISDM), settled on two configurations per model, and ran both of them on all 26. For each dataset and model, we froze whichever configuration had the lower fold-0 tune error. Each frozen configuration was then trained once on each of the five folds and scored once on that fold's test series, at the round chosen by its tune error. No test series had been seen before this point.

Table 1: The 26 MONSTER datasets. "Pool" is the number of series outside test fold 0, and "train" the rows trained on after the validation and tune slices are taken from it. Classes are those present in the labels.

dataset	series	pool	train	classes	QUANT features
AudioMNIST-DS	30,000	24,000	15,808	10	35,349
CornellWhaleChallenge	30,000	24,000	15,808	2	35,349
CrowdSourced	12,289	9,456	5,674	2	31,542
DREAMERA	170,246	140,638	132,446	2	31,542
DREAMERV	170,246	140,638	132,446	2	31,542
FordChallenge	36,257	29,003	20,811	2	14,130
FruitFlies	34,518	27,614	19,422	3	44,031
InsectSound	50,000	40,000	32,768	10	5,470
LakeIce	129,280	103,424	95,232	3	1,652
LenDB	1,244,942	983,483	975,291	2	14,940
MosquitoSound	279,566	223,652	215,460	6	33,178
Opportunity	17,386	12,184	7,312	5	117,746
PAMAP2	38,856	28,961	20,769	12	54,184
Pedestrian	189,621	151,696	143,504	82	212
S2Agri-10pc-17	5,850,881	4,661,858	4,653,666	17	2,120
S2Agri-10pc-34	5,850,881	4,661,858	4,653,666	29	2,120
STEW	28,512	21,384	13,192	2	31,542
Skoda	14,117	11,290	6,774	11	62,520
TimeSen2Crop	1,135,511	839,971	831,779	15	30,096
Tiselac	99,687	79,907	71,715	9	2,040
Traffic	1,460,968	1,168,774	1,160,582	7	212
UCIActivity	10,299	7,898	4,740	6	10,440
USCActivity	56,228	43,060	34,868	12	6,252
WISDM	17,166	11,989	7,195	6	3,126
WISDM2	149,034	98,094	89,902	6	3,126
WhaleSounds	105,163	84,130	75,938	8	22,105

Both HashBoost configurations train for up to 5,000 batches of 4,096 rows, evaluate every 50 batches, stop once their best tune evaluation is 1,000 batches old, and freeze rounds older than 128 batches:

- `G`: 6-bit gain-selected rounds ($K = 64$, so a pool of 384 candidates), $\epsilon = 100$, 8 rounds per batch. Picked on 18 datasets.
- `L`: 8-bit hard-pair rounds, $\epsilon = 1$, 16 rounds per batch. Picked on the largest pools (LenDB, both S2Agri, Traffic, Pedestrian), WhaleSounds and the EEG sets DREAMERA and DREAMERV.

Over five model seeds on fold 0, the same configuration was picked every time on 25 of the 26 datasets.

Our main baseline is XGBoost, tuned with the same effort. We chose it because it trains on the same GPU, which makes the times directly comparable, and because it can train out of core (Section 5.2). Its screen first raised the cap to 5,000 rounds with 100 rounds of patience, and then varied depth, learning rate, subsampling, L2 regularisation and the number of bins. The two final configurations are `d4` (depth 4, learning rate 0.1; picked on 17 datasets) and `d8` (depth 8, learning rate 0.05, 64 bins; picked on 9). We also compare against ExtraTrees with QUANT's own settings (200 trees, entropy criterion, a tenth of the features per split).

UCR. For smaller datasets, we use the 112 univariate UCR datasets [27] reported on by the bake-off [28]. These are the archive's 128 datasets, minus the 15 with missing values or unequal lengths, and minus Fungi, which has only one training series per class. We use each on its default train/test split, and the training sets range from 16 to 8,926 series (median 190). Since there is no validation split, every model runs at fixed settings and we report test error. HashBoost uses 800 rounds of 8 bits, ExtraTrees uses QUANT's settings, and XGBoost uses its library defaults (100 rounds, learning rate 0.3, depth 6).

Hardware and timing. Every HashBoost and XGBoost run on MONSTER had one NVIDIA L40S (46 GB) on Monash's M3 cluster to itself, so their times can be compared directly. ExtraTrees ran on CPU nodes with 36 cores. All reported times include the QUANT transform. The UCR runs and the streaming runs in Section 5.2 used one RTX 4060 (8 GB), with ExtraTrees on the same machine's 12-thread CPU. Note that seeding does not make HashBoost deterministic, because scatter-adds on the GPU accumulate in a nondeterministic order.

5 Results

5.1 Accuracy at scale (RQ1)

Table 2 gives each model's test error on the 26 MONSTER datasets, averaged over the five test folds, with training times, and Table 3 summarises the pairwise comparisons.

HashBoost matches tuned XGBoost. In geometric mean, HashBoost's error is 1.016 times XGBoost's, and it has the lower error on 14 of the 26 datasets (Table 3). The Wilcoxon test finds no difference ($p = 0.98$). A non-significant test does not on its own show that two models are equal, so we also bound the gap. A 95% t-interval over the datasets' log error ratios puts the ratio between 0.97 and 1.07, that is, within about 7% either way[1]. When one model is clearly

[1]One XGBoost run failed: WISDM's test fold 3 holds every series of two of its six classes, and XGBoost 3.4's intercept, estimated from a pool that lacks them, left it at test error 0.683. We report a rerun with the intercept fixed at 0.5, the default before version 2.0 (0.178), with the original run's time. Counting the failed run, the ratio is 0.992 ($p = 0.71$).

Table 2: MONSTER: test error, mean of the five test folds, one run per fold. HashBoost and XGBoost use the configuration each picked on fold 0. Times are training seconds summed over the five folds, QUANT included: HashBoost and XGBoost on one L40S, ExtraTrees on 36 CPU cores. The lowest error in each row is bold. †One fold rerun with a fixed intercept (footnote 1).

	HashBoost		XGBoost		ExtraTrees	time (s)		
dataset	config	error	config	error	error	HB	XGB	ET
AudioMNIST-DS	G	**0.0816**	d4	0.0884	0.1270	503	885	429
CornellWhaleChallenge	G	0.1250	d8	**0.1233**	0.1262	290	424	412
CrowdSourced	G	0.3501	d4	0.3622	**0.3396**	192	95	119
DREAMERA	L	0.5161	d8	0.5312	**0.4867**	438	1,775	4,253
DREAMERV	L	**0.4036**	d8	0.4038	0.4179	434	2,264	4,235
FordChallenge	G	0.0823	d8	0.0731	**0.0715**	196	152	224
FruitFlies	G	**0.0458**	d4	0.0462	0.0619	383	384	670
InsectSound	G	**0.1872**	d4	0.1891	0.2268	520	357	148
LakeIce	G	**0.0018**	d4	0.0019	0.0024	195	38	116
LenDB	L	0.0538	d4	**0.0522**	0.0551	728	1,364	25,843
MosquitoSound	G	0.1052	d4	**0.1008**	0.1596	1,855	5,054	7,366
Opportunity	G	0.1322	d4	**0.1305**	0.1394	399	852	761
PAMAP2	G	**0.2143**	d4	0.2246	0.2288	277	1,025	1,013
Pedestrian	L	**0.1746**	d8	0.1772	0.2135	2,588	792	61
S2Agri-10pc-17	L	0.2829	d8	0.2525	**0.2521**	1,463	30,068	18,291
S2Agri-10pc-34	L	0.2828	d8	**0.2530**	0.2638	2,013	61,933	16,151
STEW	G	**0.1937**	d4	0.1993	0.2371	325	310	382
Skoda	G	0.0567	d4	0.0610	**0.0565**	354	784	293
TimeSen2Crop	G	0.1412	d4	**0.0849**	0.1808	1,926	35,655	32,910
Tiselac	G	**0.2139**	d4	0.2140	0.2172	396	268	115
Traffic	L	0.2841	d8	0.2939	**0.2488**	846	407	375
UCIActivity	G	**0.0521**	d4	0.0675	0.0855	169	46	21
USCActivity	G	0.2428	d4	**0.2396**	0.2751	497	305	188
WISDM	G	0.1152	d4	**0.1131**†	0.1298	161	22	17
WISDM2	G	0.2556	d4	**0.2541**	0.2614	458	206	297
WhaleSounds	L	0.3128	d8	**0.2802**	0.2936	612	3,633	1,856
total						18,214	149,099	116,546

Table 3: MONSTER: pairwise comparisons over the 26 datasets. The ratio is the geometric mean of the first model's error over the second's. W/T/L counts the datasets where the first has the lower, equal and higher five-fold mean. "Clear" counts those where a 95% paired-t interval over the five per-fold differences excludes zero, either way. p is a two-sided Wilcoxon signed-rank test on the five-fold means, Holm-adjusted over the three comparisons.

comparison	ratio	W/T/L	clear (W/L)	p
HashBoost vs XGBoost	1.016	14/0/12	1/6	0.98
HashBoost vs ExtraTrees	0.904	18/0/8	9/5	0.11
XGBoost vs ExtraTrees	0.890	20/0/6	13/3	0.031

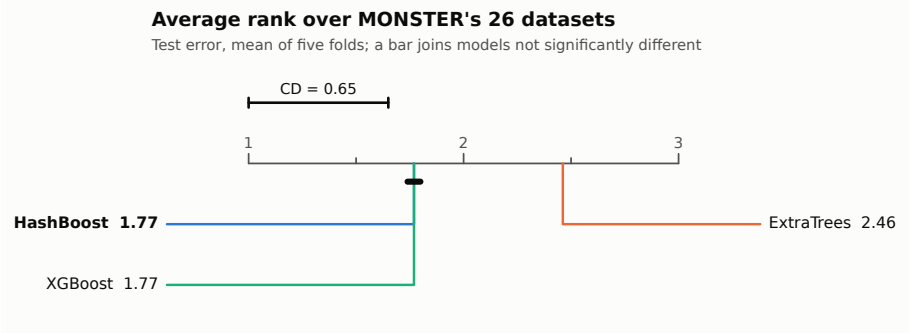


Figure 1: Critical difference diagram over MONSTER's 26 datasets [29]: test error, mean of the five folds. A Friedman test rejects equal ranks ($p = 0.016$). The bar joins models within the Nemenyi critical difference (0.65) of each other at $\alpha = 0.05$.

ahead, it is usually XGBoost, which leads clearly on 6 datasets against HashBoost's 1, among them both S2Agri subsets and TimeSen2Crop. HashBoost also does better on the 10 datasets it was developed on (ratio 0.973) than on the other 16 (1.045).

Both boosters rank ahead of ExtraTrees. Both boosters have a mean rank of 1.77, against 2.46 for ExtraTrees, which is just beyond the critical difference (Figure 1). For HashBoost the margin is narrow. Its pairwise Wilcoxon test against ExtraTrees is not significant ($p = 0.11$), although the 95% interval on its error ratio, 0.84–0.97, does exclude 1. ExtraTrees loses most on large acoustic and sensor pools such as MosquitoSound, AudioMNIST-DS and Pedestrian.

Validation error is optimistic. For all three models, test error is about 3.3 times validation error. This is because the test folds have a different distribution from the pool that the validation and tune slices are drawn from. Since every model is tuned on the same slices, the comparison is still fair, but the configurations chosen on validation error may not be the best ones on test.

5.2 Where the time goes (RQ2)

The speedup follows XGBoost's cost. On the same GPU, HashBoost trained 8.2 times faster than XGBoost in total (Table 2), but that total hides a wide range. HashBoost is faster on 14 of the 26 datasets, ranging from 31 times faster on S2Agri-10pc-34 to 7 times slower on WISDM. Figure 2 (left) shows what decides this. The speedup rises with $N \cdot p \cdot k$, the product of rows, features and classes that sets XGBoost's cost per round (Spearman $\rho = 0.87$), whereas HashBoost's cost is set by its batch budget. On the cheapest datasets, such as WISDM, LakeIce and UCIActivity, XGBoost finishes in seconds, and HashBoost's fixed cost per round makes it the slower of the two. On the most expensive ones, S2Agri and TimeSen2Crop, XGBoost takes hours while HashBoost takes minutes.

At equal time, XGBoost is far less accurate on the expensive datasets. In Figure 2 (right), XGBoost gets exactly HashBoost's total training time on each fold, patience included, and its round is chosen on tune error from the rounds it managed to complete. Under this limit HashBoost is ahead on validation error on 19 of the 26 datasets. Across all 26 the difference is not significant (Wilcoxon $p = 0.24$), because on the cheaper datasets XGBoost finishes within HashBoost's time anyway. Where training is expensive, however, the gap is large. Given HashBoost's time, XGBoost only reaches 0.27–0.29 on the two S2Agri subsets, against HashBoost's 0.02–0.03, and the gap on TimeSen2Crop is similar. Figure 3 shows the curves for the six largest pools.

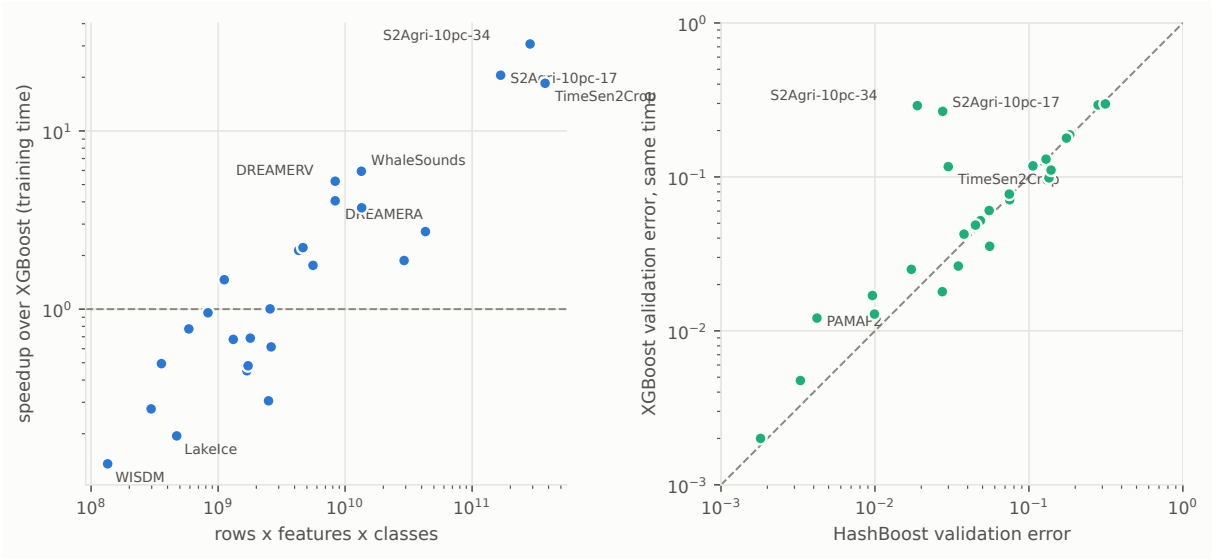


Figure 2: Left: HashBoost's training speedup over XGBoost on each MONSTER dataset (time summed over the five folds, one L40S) against rows × features × classes. Above the dashed line, HashBoost is faster. Right: validation error when XGBoost is given the same training time as HashBoost, mean of the five folds. XGBoost's round is chosen on tune error among the rounds it completed in that time. Above the diagonal, HashBoost is ahead at equal time.

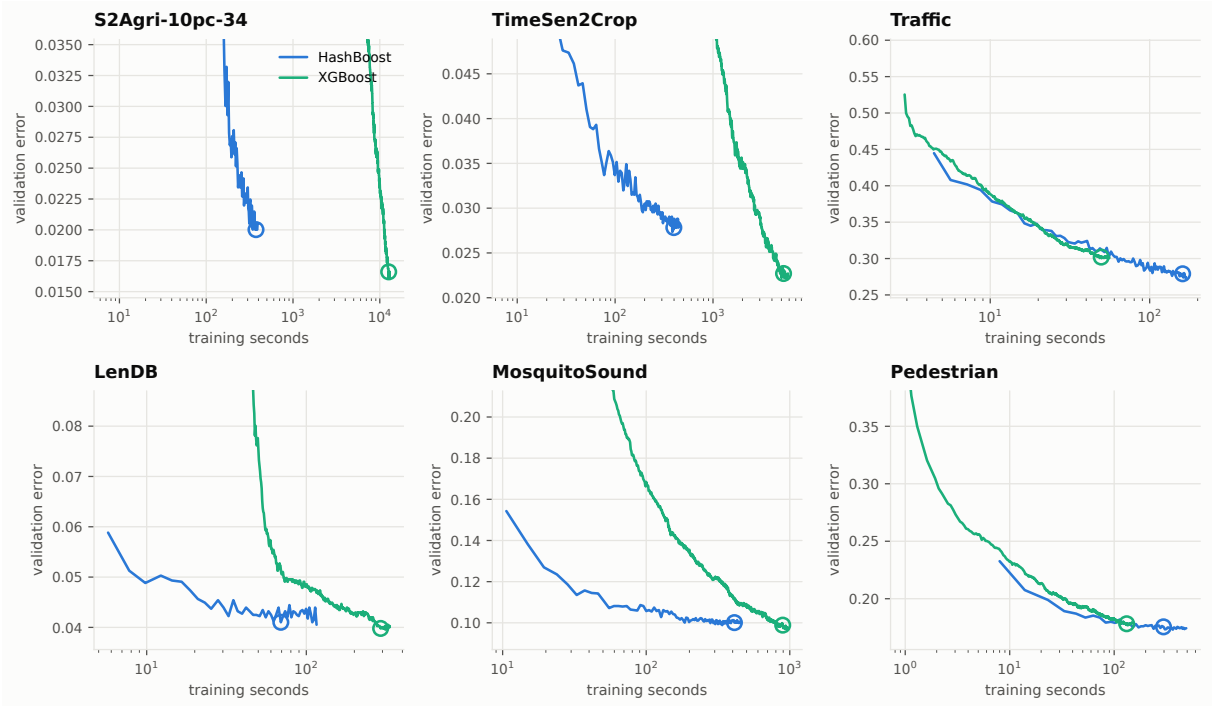


Figure 3: Validation error against training time on fold 0 for the six largest MONSTER pools, each booster at its frozen configuration, on one L40S, QUANT included. HashBoost's time is measured at each evaluation; XGBoost's is its build plus its mean time per round. Rings mark the tune-chosen point. The horizontal axis is logarithmic.

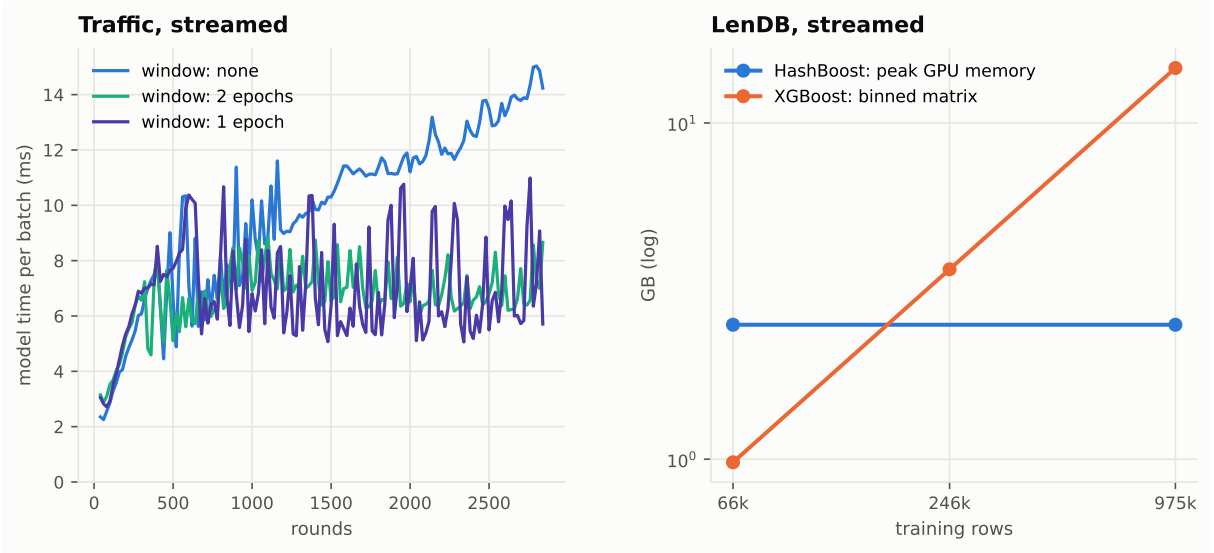


Figure 4: Left: HashBoost's model time per batch against rounds on a whole-pool Traffic stream (1.16 million rows, one round per batch, RTX 4060), unfrozen and with rounds older than one or two epochs frozen. Right: what each model needs available as LenDB's training pool grows. HashBoost's peak GPU memory is measured; XGBoost's binned matrix is measured on disk at 245,760 and 975,291 rows and derived from the same bytes per element at 65,536.

HashBoost's budget ran out on the largest pools. When it is run to completion, XGBoost overtakes HashBoost on S2Agri and TimeSen2Crop. HashBoost, however, was stopped by its budget rather than by its tune error. On 14 of the 15 S2Agri and Traffic folds it reached its 5,000-batch cap with its best tune error in the last 1,000 batches, which means it was still improving. On TimeSen2Crop it reached the cap on only two of the five folds.

Memory does not grow with the training set. We streamed LenDB's whole training pool (975,291 rows, which is 6.3 GB as raw series and 58 GB after QUANT) through both models on the RTX 4060. HashBoost's peak GPU memory was 2,572 MB both at 65,536 rows and at the full pool, because each batch is folded into the tables and then freed. XGBoost's binned matrix grows linearly with the pool, reaching 14.6 GB (Figure 4, right). On the 8 GB card, XGBoost had to page that matrix in from disk every round, which took 116 s per round, compared with 0.68 s at 65,536 rows. In the MONSTER benchmark on the L40S, tuned XGBoost used about 17 GB of GPU memory on LenDB, while HashBoost used at most 4.9 GB.

Freezing keeps the cost per batch flat. Without freezing, the cost per batch grows with the number of rounds. On a streamed Traffic run it rose from 2 ms to 14.5 ms over 2,840 rounds, whereas a window of one or two epochs' worth of rounds kept it level (Figure 4, left). On Pedestrian, with two rounds per batch, a window of ten epochs matched the unfrozen model's validation error in 2.4 times less time.

5.3 The small-data boundary (RQ3)

The UCR 112 training sets are small, and on them HashBoost loses both its accuracy and its speed advantage (Table 4). Our ExtraTrees reproduces the published QUANT results [28]. It is the most accurate model, significantly ahead of both boosters (Figure 5), and also the fastest, taking under half of HashBoost's time. At these sizes, most of a round's cost is fixed overhead rather than arithmetic. With its MONSTER defaults, HashBoost is level with default XGBoost. Its 8-bit rounds were set for buckets that see thousands of rows, and most UCR training sets are too small to fill them. When we resize it for small data, using two bits and 3,200 rounds chosen

Table 4: UCR 112: test error on the default splits, one run per dataset. Mean rank is among the three models we ran. W/T/L is HashBoost's record against each row. p is a two-sided Wilcoxon signed-rank test against HashBoost, Holm-adjusted over the four comparisons. The published rows are the bake-off's resample 0, which is the default split.

model	mean error	median	mean rank	W/T/L	p	total fit
HashBoost	0.1941	0.1705	2.36	–	–	183 s
ExtraTrees	**0.1475**	0.1205	**1.25**	8/8/96	2.6×10^{-14}	70 s
XGBoost	0.2101	0.1912	2.39	58/3/51	0.082	413 s
QUANT (published)	0.1456	0.1177		13/5/94	4.9×10^{-14}	
HC2 (published)	0.1236	0.0753		8/7/97	3.8×10^{-15}	

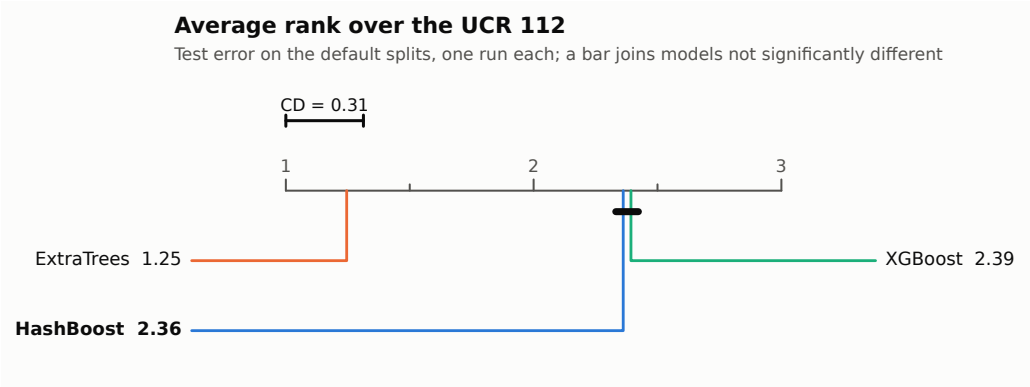


Figure 5: Critical difference diagram over the UCR 112 [29]. A Friedman test rejects equal ranks ($p = 3.6 \times 10^{-22}$). A bar joins models within the Nemenyi critical difference (0.31) of each other at $\alpha = 0.05$.

by two-fold cross-validation over the datasets, HashBoost is significantly ahead of XGBoost (lower error on 82 datasets against 29), but still behind ExtraTrees (28 against 63), and it takes 7.5 times its default training time. So the best classifier for QUANT's features depends on scale. An average of random trees works best on small training sets, and a booster works best on large ones.

6 Discussion

Speed against accuracy. The two models' costs scale differently, so the speedup is largest on the datasets where the classifier's cost was a problem in the first place, and those are the datasets HashBoost is meant for. They are also where it gives up the most accuracy once XGBoost has been run to completion: its error is 1.12 times XGBoost's on S2Agri and 1.66 times on TimeSen2Crop, even though it is far ahead at equal time. On S2Agri, the trade-off comes at least partly from a budget that was tuned on smaller pools, rather than from a limit of the model itself. On TimeSen2Crop the budget explains less of the gap, which is the largest in the benchmark.

Limitations.

- The MONSTER benchmark is one run per test fold. Over five model seeds on fold 0, HashBoost's standard deviation was 0.0002 to 0.0024 on most datasets.
- Both models were tuned for accuracy, not speed, and only on 10 of the 26 datasets, which flatter HashBoost (Section 5.1).

- The time-matched comparison is on validation error, because test error was scored only at each model's own chosen round. XGBoost's clock is its build plus its mean time per round.
- The UCR results are one run on the default split, where the bake-off averages 30 resamples.
- The comparison includes XGBoost and ExtraTrees only, not LightGBM or CatBoost.

Future work. The results on the largest datasets suggest that a round budget that grows with the training pool would gain accuracy for a few more minutes of training. A configuration tuned for speed rather than accuracy would show the trade-off on the test folds. Choosing the bit width from the number of rows per class might unify the UCR and MONSTER settings, and cheaper rounds would help on small datasets.

7 Conclusion

Gradient boosting can be accurate at scale without trees and without searching the training set. A global partition placed between hard examples, with its leaves refined by every later mini-batch, matched tuned XGBoost on MONSTER, at a cost set by its own budget rather than by the size of the data. As a result it has the biggest advantage on the datasets where tree boosting is slowest. On small datasets it is the wrong choice, since ExtraTrees is both faster and more accurate there.

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